Data Investigations

Establishing a process for conducting a disciplined forensic capability around data issues.

Data Investigations Overview

- **Data Investigations (DI) is brought in to provide forensics, analysis, and direct remediation for addressing data issues.**

Remediation

  - Note: DI is not the first line of defense, nor is it necessarily customer facing (though they may be asked to participate in discussions with customers and/or 3rd parties).

- **DI is responsible for directing research on data issues to identify scope, impact, and to guide Causal Factor Analysis (root cause).**

Prevention

DI Team Responsibilities

- **Call Center & Help Desk**
  - Ensure data issues are consistently captured for traceability and impact analysis (whether discovered internally or externally to the Org.).
  - Ensure an escalation plan is in place to involve the DI team when and where appropriate (*DI Team to provide guidance*).
- **DI Team (reactive)**
  - Capture all information necessary to perform Causal Factor Analysis (CFA) when a data issues is discovered.
  - Engage technical resources (tools and/or individuals) to perform forensic research (within the Org. and beyond the Org.).
  - Document all activities performed during an investigation to support audit reporting needs.
  - Collaborate with associate teams in the DG Operations Team (typically) to identify options and propose a recommendation for remediating the data issue (short term to address the immediate problem & long term to prevent a reoccurrence of the problem).

DI Responsibilities

- **DI Team (proactive)**
  - Identify research opportunities based on analysis of existing or potential risk involving data (internal and/or external).
  - Conduct research to identify causal factors for data issues
  - Collaborate with associate teams in the DG Operations Team (typically) to identify options and propose a recommendation for remediating the data issue (short term to address the immediate problem & long term to prevent a reoccurrence of the problem).
- **Information Architecture Team**
  - Typically, the Info Architecture Team will get involved in the proactive efforts to identify options for addressing risk BEFORE it happens.
  - The Info Architecture Team may get involved with defining longer term solutions to prevent problems from recurring by identifying architectural deficiencies with structures (schemas) or systems (tools or processes).

DI Responsibilities

- **Quality Assurance Team**
  - Ensure any short term “fixes” that are implemented are properly documented as such with follow on plans to replace short term fixes (e.g. work around, system patch, data modification, …) with appropriate solutions which follow proper change control procedures.
  - Ensure appropriate controls are in place when data modifications are executed. (e.g. Production Data should never be altered directly)
- **Business SMEs Team**
  - Ensure any changes impacting the Org. systems (or non-Org systems – 3rd parties) are identified and communicated in advance to the DI Team, operations, and 3rd parties.
  - Coordinate communications and remediation plans with the DI Team to minimize disruption to all parties (esp. Customers).
  - Collaborate with the QA Team, DI Team, and ADB staff (where appropriate) to ensure short term fixes are properly implemented (order, timing, scope, accuracy, with minimized impact to ongoing operations).

DI Team Research Pipeline

- The following list of items *may* be useful for the DI Team to consider for proactively improving data quality within the Org. (primary) without the Org. (secondary):

  - Research: When & Why do are Data Mods (CFMs) required ?
  - Research: How should Data Forensics be performed in the Org. ?
  - Research: What types (classes) of Data Issues exist ?
  - Research: What is the quality of data received from the Org. Vendors ?
  - Research: What is the quality of data delivered to the Org. Customers ?
  - Research: What is the quality of data delivered to the Org. Partners ?

SAMPLE DELIVERABLES

Sample Deliverables from DI

Types of Analysis

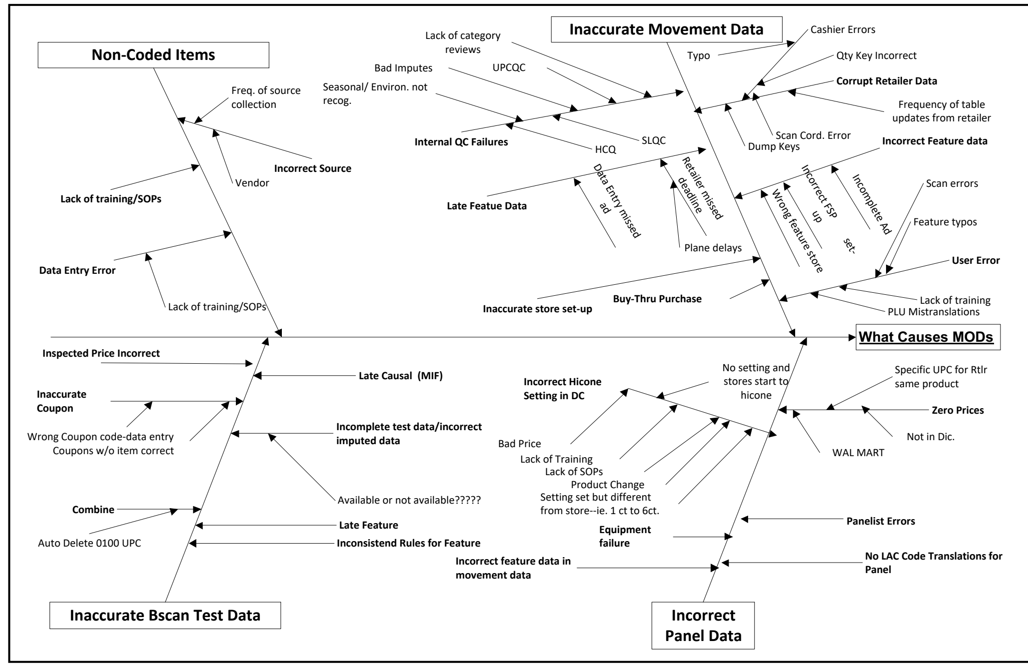
- ❑ **Process Flow Maps (current state)**
  - Processes involved in the generation of a data issue
- ❑ **Customer Focus Study**
  - Studies internal customers and external customers which may provide insights on why data issues are occurring
- ❑ **Cause & Effect Diagrams**
  - Fishbone diagrams (see below) can reveal patterns and more efficiently direct more focused study
- ❑ **Data Collection Plan**
  - Defines what data needs to be captured for the analysis (location, amount, sources, and how it needs to be extracted & captured)
- ❑ **Pareto Charts (see samples below)**

Sample Deliverables from DI

Cause & Effect Diagrams

- Fishbone analysis can identify numerous sources for causality.

- Alignment of causality can reveal patterns and lead to better solutions.



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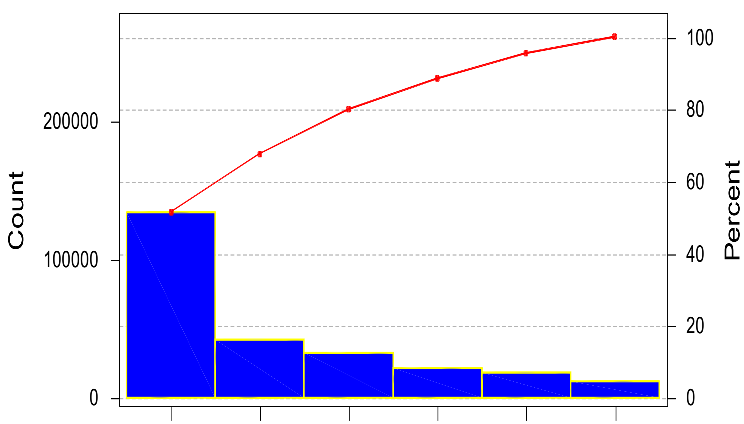
Sample Deliverables from DI

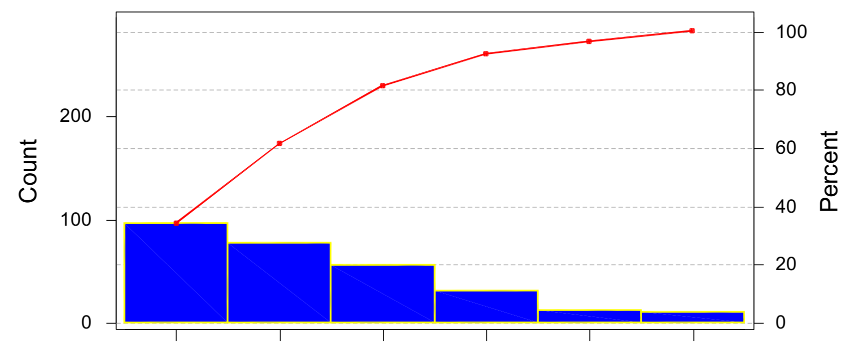
Control Charts Comparing Requested Mods vs. Applied Mods

Mods applied



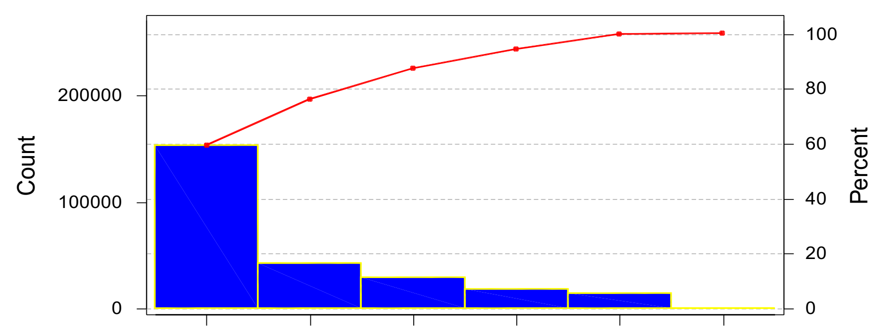
Defect	PD	FO	TS	DQ	PC	Others
Count	134142	41979	32409	21712	18393	11473
Percent	51.6	16.1	12.5	8.3	7.1	4.4
Cum %	51.6	67.7	80.2	88.5	95.6	100.0

Requests sent (number of letters)



Defect	DQ	TS	IS	PD	PC	Others
Count	96	77	56	31	12	10
Percent	34.0	27.3	19.9	11.0	4.3	3.5
Cum %	34.0	61.3	81.2	92.2	96.5	100.0

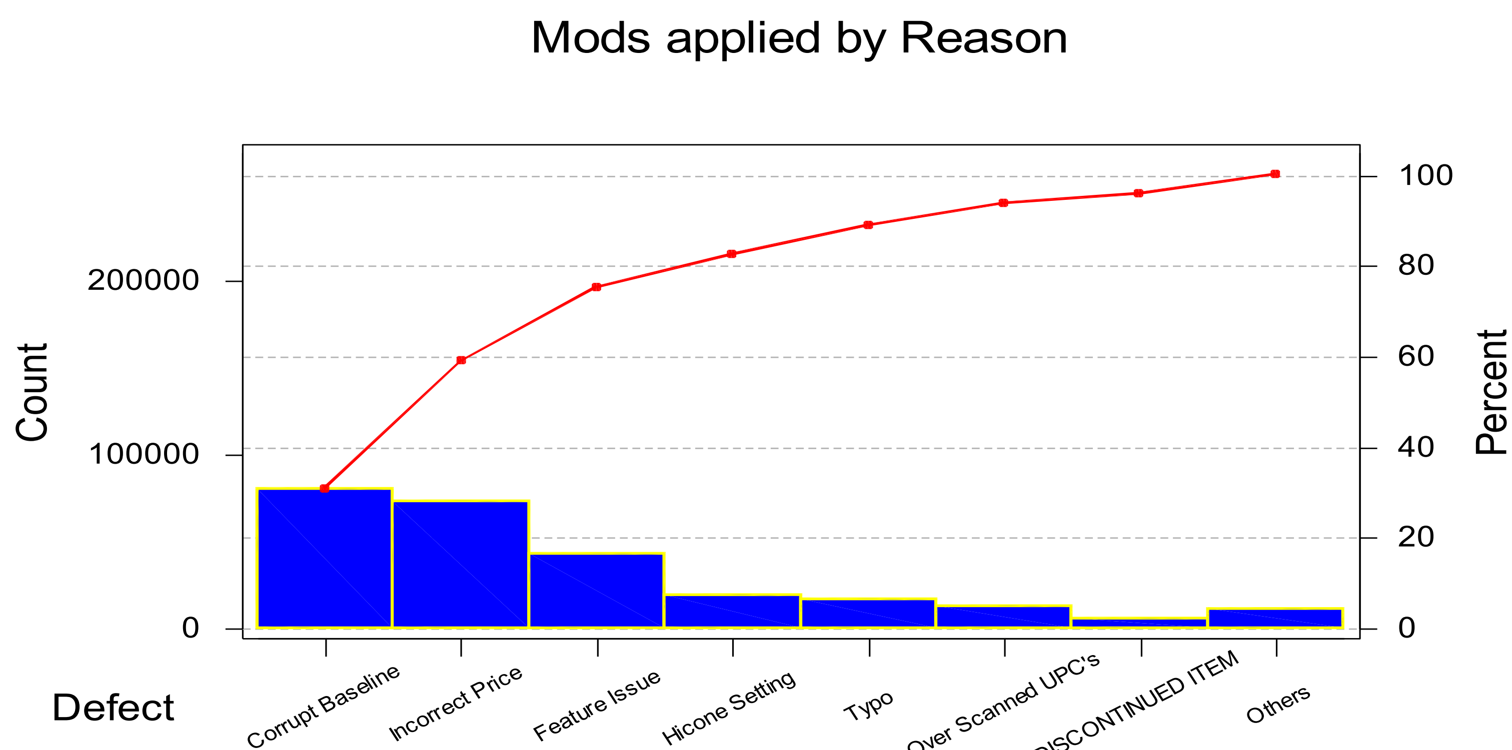
Mods requested



Defect	PD	FO	DQ	TS	PC	Others
Count	153154	42524	29199	18235	14051	371
Percent	59.5	16.5	11.3	7.1	5.5	0.1
Cum %	59.5	76.0	87.3	94.4	99.9	100.0

Sample Deliverables from DI

Control Charts citing reasons MODS are requested



Defect	Corrupt Baseline	Incorrect Price	Feature Issue	Hicone Setting	Typo	Over Scanned UPC's	DISCONTINUED ITEM	Others
Count	80105	73159	42419	18680	16469	12904	5673	10699
Percent	30.8	28.1	16.3	7.2	6.3	5.0	2.2	4.1
Cum %	30.8	58.9	75.2	82.4	88.7	93.7	95.9	100.0

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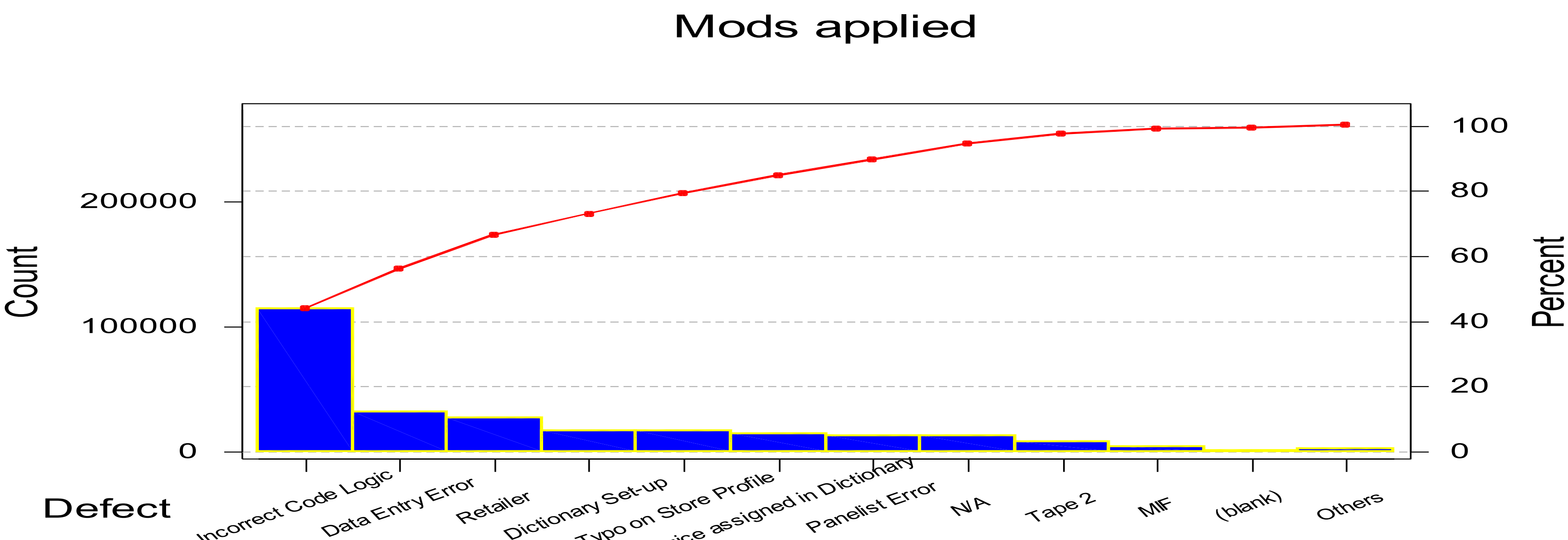
Sample Deliverables from DI

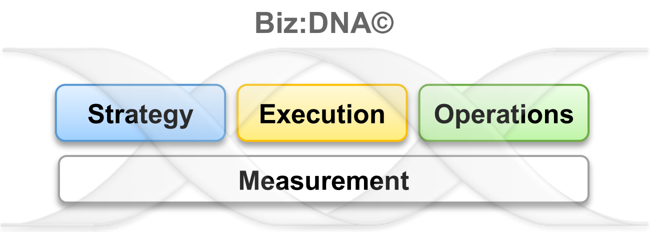
Baseline Analysis showing causes for Mods to be applied



Defect	Incorrect Code Logic	Data Entry Error	Retailer	Dictionary Set-up	Typo on Store Profile	No price assigned in Dictionary	Panelist Error	N/A	Tape 2	MIF	(blank)	Others
Count	113692	31791	27164	16915	16469	13993	12988	12299	7852	3975	859	2111
Percent	43.7	12.2	10.4	6.5	6.3	5.4	5.0	4.7	3.0	1.5	0.3	0.8
Cum %	43.7	55.9	66.4	72.9	79.2	84.6	89.6	94.3	97.3	98.9	99.2	100.0

Thank You

Measure what matters
Create and deliver the right value
Build and nurture a winning culture
Proactively manage change & complexity



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